

Problem 1 (Munkres 28.3). Let X be limit point compact.

- (a) If $f: X \rightarrow Y$ is continuous, does it follow that $f(X)$ is limit point compact?
- (b) If A is a closed subset of X , does it follow that A is limit point compact?
- (c) If X is a subspace of the Hausdorff space Z , does it follow that X is closed in Z ?

Solution. (a) No. Let Y be a two-point set with the indiscrete topology and let $X = \mathbb{N} \times Y$, as in Section 28 Example 1. Then, as Munkres shows in the example, X is limit point compact, but its image under the projection $\pi_1: X \rightarrow \mathbb{N}$ is $\mathbb{N}$, which is not limit point compact because it is infinite but every point is an isolated point and so no subset has a limit point.

(b) Yes. If $E \subseteq A$ is infinite, then it is an infinite subset of X , so E has a limit point x . This point x is a limit point of A as well, which means $x \in A$ because closed sets contain their limit points, which means that E has a limit point in A ; this proves that A is limit point compact.

(c) No. Let ω_1 be the first uncountable ordinal (S_Ω in Munkres's language, but I will use the standard notation). Then $\omega_1 + 1 = [0, \omega_1]$ is Hausdorff in the order topology, and the subspace $[0, \omega_1) = \omega_1$ is limit point compact because it is sequentially compact: given countable ordinals $\alpha_1, \alpha_2, \dots$, their supremum (union) $\alpha = \bigcup_{n \in \mathbb{N}} \alpha_n$ is a countable ordinal because unions of countably many countable sets are countable, so $\alpha \in \omega_1$, and ω is a limit point of the sequence because if there were an open interval around α not containing any α_n then we would be able to take the smaller endpoint of the interval to get a smaller upper bound for the sequence. To show that ω_1 is not closed in $\omega_1 + 1$, note first that $\omega_1 + 1$ is compact by Munkres's Theorem 27.1, where $[0, \omega_1]$ has the least upper bound property because the least upper bound of a set of ordinals is their union. On the other hand $[0, \omega_1)$ is not compact because the sets of the form $[0, \alpha)$ where $\alpha \in \omega_1$ form an open cover with no finite subcover. Closed subspaces of compact spaces are compact, so if $[0, \omega_1)$ were closed then it would be compact; therefore it is not closed in $[0, \omega_1]$. $\square$

Problem 2 (Munkres 28.6). Let (X, d) be a metric space. If $f: X \rightarrow X$ satisfies the condition $d(f(x), f(y)) = d(x, y)$ for all $x, y \in X$, then f is called an isometry of X . Show that if f is an isometry and X is compact, then f is bijective and hence a homeomorphism.

Solution. f is injective because

$$f(x) = f(y) \implies 0 = d(f(x), f(y)) = d(x, y) \implies x = y.$$

Now suppose that f is not surjective, and let $a \in X \setminus f(X)$. Since for any $\varepsilon > 0$, $d(x, y) < \varepsilon$ implies $d(f(x), f(y)) < \varepsilon$, f is continuous, so $f(X)$ is compact, hence closed in X (since the metric space X is Hausdorff). This means we can find $\varepsilon > 0$ such that $B_\varepsilon(a)$ is disjoint from $f(X)$. Then, define the sequence $x_0 = a$ and $x_n = f(x_{n-1})$ for $n \in \mathbb{N}$. Define $f^1 = f$ and $f^{n+1} = f^n \circ f$ for all $n \in \mathbb{N}$. For any positive integers $m > n$, we have

$$d(x_m, x_n) = d(f^n(f^{m-n}(a)), f^n(a)) = d(f^{m-n}(a), a) \geq \varepsilon,$$

where the second equality comes from f , hence the iterated function f^n , being an isometry, while the inequality comes from $f^{m-n}(a)$ being in $f(X)$ (so it cannot lie in $B_\varepsilon(a)$ by assumption). However,

since X is compact, the sequence $x_1, x_2, \dots$ has a convergent subsequence, which is a Cauchy sequence, so there exist two points in the subsequence that are less than ε away from each other, contradicting the inequality that we just proved. Therefore, f must be surjective, so together with injectivity we have that f is a bijection on X .

We also already established that f (hence any isometry) is continuous, and f^{-1} must therefore be continuous because it is also an isometry, since if $x, y \in X$ then

$$d(x, y) = d(f(f^{-1}(x)), f(f^{-1}(y))) = d(f^{-1}(x), f^{-1}(y))$$

where the second equality is due to f being an isometry. This proves that f is a continuous bijection with continuous inverse, so it is a homeomorphism of X . $\square$

Problem 3 (Munkres 28.7). Let (X, d) be a metric space. If $f: X \rightarrow Y$ satisfies the condition $d(f(x), f(y)) < d(x, y)$ for all $x, y \in X$ with $x \neq y$, then f is called a **shrinking map**. If there is a number $\alpha < 1$ such that

$$d(f(x), f(y)) \leq \alpha d(x, y)$$

for all $x, y \in X$, then f is a **contraction**.

- (a) If f is a contraction and X is compact, show that f has a unique fixed point.
- (b) Show more generally that if f is a shrinking map and X is compact, then f has a unique fixed point.
- (c) Let $X = [0, 1]$. Show that $f(x) = x - x^2/2$ maps X into X and is a shrinking map that is not a contraction.
- (d) Show that the map $f: \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = (x + \sqrt{x^2 + 1})/2$ is a shrinking map that is not a contraction and has no fixed point.

Solution. (a) Let $A_n = f^n(X) = \{f^n(x) : x \in X\}$, with $A_0 = X$. If $y \in A_n$ then $y = f^n(x) = f^{n-1}(f(x))$ for some $x \in X$ so $y \in A_{n-1}$ as well, meaning $A_{n-1} \subseteq A_n$ for all $n \in \mathbb{N}$. Since X is compact, it is bounded in the metric d , so there exists some $C > 0$ such that $d(x, y) < C$ for all $x, y \in X$. Then, we can see inductively that the diameter of A_n is less than $\alpha^n C$: the base case A_0 is that the diameter of X is less than C and, if A_n has diameter less than $\alpha^n C$, then for any $x, y \in A_{n+1}$, if $x = f(x')$ and $y = f(y')$ for $x', y' \in A_n$,

$$d(x, y) = d(f(x'), f(y')) \leq \alpha d(x', y') < \alpha \cdot \alpha^n C = \alpha^{n+1} C.$$

Since X is compact and each A_n is compact, hence closed, because it is the image of compact set X , this implies that $A = \bigcap_{n=1}^{\infty} A_n$ is nonempty. The diameter of A is bounded above by $\alpha^n C$ for each positive integer n , so A has diameter zero, meaning it consists of a single point p . Any fixed point of f must be contained in A because if $x \in A_n$ then $f(x) = x \in A_{n+1}$, so this shows that if p is a fixed point it is unique. To see that p is indeed a fixed point, note that $f(A)$ is a subset of $\bigcap_{n=1}^{\infty} f(A_n) = \bigcap_{n=2}^{\infty} A_n = A$, and $f(A)$ must be nonempty because A is nonempty, so $f(A)$ is equal to A , the only nonempty subset of A , meaning f fixes p .

(b) Define A_n and A as above. As in the proof above, $f(A)$ is a subset of A . We show that A is a subset of $f(A)$. To do this let $a \in A$, and for each n pick $x_n \in X$ such that $f^{n+1}(x_n) = a$; such x_n exists because $a \in A_{n+1}$. Then, the sequence $y_n = f^n(x_n), \dots$ has a convergent subsequence $(y_{n_k})_{k=1}^{\infty}$, say with limit y , because X is compact, hence sequentially compact. If f is a shrinking map it must be continuous because for any $\varepsilon > 0$, $d(x, y) < \varepsilon$ implies $d(f(x), f(y)) < d(x, y) < \varepsilon$, and continuity in a metric space is the same thing as sequential continuity, so

$$a = \lim_{n \rightarrow \infty} f^{n+1}(x_n) = \lim_{k \rightarrow \infty} f(y_{n_k}) = f(\lim_{k \rightarrow \infty} y_{n_k}) = f(y)$$

(since any subsequence of a convergent sequence has the same limit as the parent sequence); thus $a \in f(A)$. This implies that A and $f(A)$ are equal, as they contain each other. Noting that A is compact (because it is the intersection of compact sets), $A \times A$ is compact as well, so the continuous function $d: A \times A \rightarrow [0, \infty)$ achieves its maximum for some pair of points $(x, y) \in A \times A$; this maximum value is by definition the diameter of A . Similarly, there exist points $x', y' \in f(A)$ such that $d(x', y')$ is the diameter of $f(A)$ (because $f(A)$ is the continuous image of a compact set, hence compact). By proving $A = f(A)$ we have shown that A and $f(A)$ have the same diameter; if this diameter were nonzero then, picking $x_0, y_0 \in A$ such that $f(x_0) = x'$ and $f(y_0) = y'$, we have

$$\text{diam } A \geq d(x_0, y_0) \geq d(f(x_0), f(y_0)) \geq d(x', y') = \text{diam } f(A),$$

so equality must hold in the inequality above; in particular $d(x_0, y_0) = d(f(x_0), f(y_0))$ can hold if and only if $x_0 = y_0$, so that $A = f(A)$ consists of exactly one point; the element of A is a fixed point, and its uniqueness follows via the same logic as in part (a), where any fixed point of f would be contained in all the A_n , hence also in A .

(c) f is a shrinking map because for any distinct $x, y \in [0, 1]$,

$$|f(x) - f(y)| = |x - y - x^2/2 + y^2/2| = |(x - y)(1 - x/2 - y/2)| \leq |x - y| \cdot \left| 1 - \frac{x + y}{2} \right|$$

and $0 \leq \frac{x+y}{2} \leq 1$ so $1 - \frac{x+y}{2}$ is between 0 and 1 as well, hence so is its absolute value; thus

$$|f(x) - f(y)| < |x - y|.$$

On the other hand it is not a contraction because

$$1 = f'(0) = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h},$$

so for each $c < 1$ there is an $h > 0$ such that $f(h) - f(0) > ch$ — to be more precise, if f were a contraction then $\sup_{h \in (0, 1]} \frac{f(h) - f(0)}{h - 0}$ would be less than 1, but the limit above equalling 1 implies that this supremum is at least 1, so f cannot be a contraction.

(d) f has no fixed point because

$$f(x) = \frac{x + \sqrt{x^2 + 1}}{2} > \frac{x + \sqrt{x^2}}{2} \geq \frac{2x}{2} = x$$

for all x . To show that it is a shrinking map, we start by writing

$$|f(x) - f(y)| \leq \left| \frac{x - y}{2} \right| + \left| \frac{\sqrt{x^2 + 1} - \sqrt{y^2 + 1}}{2} \right|.$$

Since the derivative of $\frac{\sqrt{t^2 + 1}}{2}$ is $\frac{t}{2\sqrt{t^2 + 1}}$, there exists ξ between x and y such that the latter addend is equal to

$$|x - y| \cdot \frac{\xi}{2\sqrt{\xi^2 + 1}} < \frac{|x - y|}{2}$$

by the mean value theorem; this proves that $|f(x) - f(y)| < |x - y|$.

To show that f is not a contraction we note that

$$\lim_{x \rightarrow \infty} \frac{|f(x) - f(0)|}{|x - 0|} = \lim_{x \rightarrow \infty} \left| \frac{x + \sqrt{x^2 + 1} - 1}{2 \cdot x} \right| = \lim_{x \rightarrow \infty} \left| \frac{1}{2} + \frac{1}{2} \sqrt{1 + \frac{1}{x^2}} - \frac{1}{2x} \right| = 1$$

meaning that for any $c < 1$, there exists $N_c \in \mathbb{R}$ such that $x > N_c$ implies $\frac{|f(x) - f(0)|}{|x - 0|} > c$. If f were a contraction then there would exist $\alpha < 1$ such that $|f(x) - f(0)| \leq \alpha |x - 0|$, but taking $x > N_\alpha$ shows that the reverse inequality holds, so f is not a contraction. $\square$

Problem 4 (Munkres 29.1). Show that the rationals $\mathbb{Q}$ are not locally compact.

Solution. Let $x \in \mathbb{Q}$ be an arbitrary point and let $C \subseteq \mathbb{Q}$ be a set containing an open set U around x . If C were compact, then it would be sequentially compact by Munkres's Theorem 28.2. However, we can construct a sequence of points in U that does not converge in C : let $(a, b) \cap \mathbb{Q}$ be a basis element of the subspace topology on $\mathbb{Q}$ that is contained in U and let $r \in (a, b)$ be irrational. Then, since $\mathbb{Q}$ is dense in $\mathbb{R}$, we can find for each $n \in \mathbb{N}$ a rational number $r_n \in (a, b) \cap \mathbb{Q}$ such that $|r - r_n| < \frac{1}{n}$, so that $\lim_{n \rightarrow \infty} r_n = r$, which implies any subsequence also converges to r , so in particular no subsequence of r_n converges to a point in $\mathbb{Q}$ (since limits of sequences are unique in Hausdorff spaces). Thus there does not exist a compact subset of $\mathbb{Q}$ that contains an open neighborhood of x , so $\mathbb{Q}$ is not locally compact. $\square$

Problem 5. Show that the one-point compactification of $\mathbb{R}$ is homeomorphic with the circle S^1 .

Solution. Denote by $\overline{\mathbb{R}}$ the one-point compactification of $\mathbb{R}$ and let ω be the point of $\overline{\mathbb{R}} \setminus \mathbb{R}$. Define a **wrapped interval** $a, b($ as $(-\infty, a) \cup (b, \infty) \cup \{\omega\}$, where a, b are real numbers such that $a < b$.

I claim that

$$\mathcal{B} = \{(a, b) : a, b \in \mathbb{R}, a < b\} \cup \{a, b(: a, b \in \mathbb{R}, a < b\}$$

is a basis for the topology on $\overline{\mathbb{R}}$. The topology on $\overline{\mathbb{R}}$ consists of two kinds of sets:

- (1) the subsets of $\mathbb{R}$ that are open in $\mathbb{R}$; and
- (2) complements in $\overline{\mathbb{R}}$ of compact subsets of $\mathbb{R}$ (so sets of this kind contain ω).

By definition, a wrapped interval is $\overline{\mathbb{R}} \setminus [a, b]$ where a, b are reals with $a < b$, so elements of $\mathcal{B}$ are clearly open in $\overline{\mathbb{R}}$. Conversely, we show that open subsets of $\overline{\mathbb{R}}$ are in the topology generated by $\mathcal{B}$. Sets of the first kind are in the topology generated by $\mathcal{B}$ because the open intervals are in $\mathcal{B}$. To show that sets of the second kind are in $\mathcal{B}$, note that if C is a compact subset of $\mathbb{R}$, then C is bounded by the Heine-Borel theorem, so C can be contained in $[a, b]$ for some real numbers a, b with $a < b$; picking $\varepsilon > 0$ arbitrarily, we can say that

$$\overline{\mathbb{R}} \setminus C = (\overline{\mathbb{R}} \setminus [a, b]) \cup \underbrace{((a - \varepsilon, b + \varepsilon) \setminus C)}_{U \text{ open in } \mathbb{R}} = a, b(\cup U$$

so sets of the second kind are unions of open sets in $\mathbb{R}$ with a wrapped interval, hence open in the topology generated by $\mathcal{B}$.

Let S^1 be the unit circle in $\mathbb{R}^2$ and for $\theta \in \mathbb{R}$, identify the point $(\cos \theta, \sin \theta) \in S^1$ with θ . Define $f: S^1 \rightarrow \overline{\mathbb{R}}$ by

$$f(0, 1) = \omega, \quad f(\cos \theta, \sin \theta) = \frac{\cos \theta}{1 - \sin \theta} = \tan \left(\frac{\theta}{2} + \frac{\pi}{4} \right) \text{ if } \sin \theta \neq 1,$$

where the expression for f comes from

$$\frac{\cos \theta}{1 - \sin \theta} = \frac{\sin(\pi/2 - \theta)}{1 - \cos(\pi/2 - \theta)} = \frac{2 \sin(\pi/4 - \theta/2) \cos(\pi/4 - \theta/2)}{2 \sin^2(\pi/4 - \theta/2)} = \cot \left(\frac{\pi}{4} - \frac{\theta}{2} \right) = \tan \left(\frac{\theta}{2} + \frac{\pi}{4} \right).$$

Since the value of f increases as θ increases over $(-3\pi/2, \pi/2)$ and the restricted map $\widehat{f}: S^1 \setminus \{0, 1\} \rightarrow \mathbb{R}$ is continuous (because it is a quotient of continuous functions), f maps open arcs in S^1 not containing ω to open intervals in $\mathbb{R}$. Furthermore, open arcs in S^1 containing $(0, 1)$ are really unions of two open arcs, each with one endpoint at $(0, 1)$, and $\pi/2$. If $3\pi/2 < \alpha < \beta < \pi/2$ are real numbers then the counterclockwise arc from $(\cos \beta, \sin \beta)$ to $(\cos \alpha, \sin \alpha)$ contains $(0, 1)$, so can be decomposed into the counterclockwise arc from $(\cos \beta, \sin \beta)$ to $(0, 1)$, the counterclockwise arc from $(0, 1)$ to $(\cos \alpha, \sin \alpha)$, and the point $(0, 1)$. Since these two arcs do not contain $(0, 1)$ they get mapped by f to open intervals, where the fact that f increases with the angles of the points means

that the former arc maps to $(f(\cos \beta, \sin \beta), \infty)$ and the latter arc to $(-\infty, f(\cos \alpha, \sin \alpha))$. This means f sends arcs containing $(0, 1)$ to wrapped intervals, as the two intervals above together with $f(0, 1) = \omega$ constitute a wrapped interval. This proves that f is an open map, as basis elements of the topology on S^1 map to basis elements of the topology on $\overline{\mathbb{R}}$ so that open sets get sent to open sets.

Since $\tan(\theta/2 + \pi/4)$ is bijective on $(-3\pi/2, \pi/2)$, $\widehat{f}$ is bijective, so f is also bijective (the one extra point $(0, 1)$ in the domain of f maps to the one extra point ω in the codomain).

Let $g: (-3\pi/2, \pi/2) \rightarrow \mathbb{R}$ be the map $g(\theta) = \tan(\frac{\theta}{2} + \frac{\pi}{4})$; this is bijective. The inverse g^{-1} is increasing, and $f^{-1}(r) = (\cos(g^{-1}(r)), \sin(g^{-1}(r)))$ when $r \neq \omega$, and of course $f^{-1}(\omega) = (0, 1)$. The increasing nature of g means that open intervals in $\mathbb{R}$ map by f^{-1} to open arcs in S^1 not containing $(0, 1)$, with unbounded open intervals mapping to open intervals having $(0, 1)$ as an endpoint. By decomposing wrapped intervals into unions of intervals with $\{\omega\}$, we can see that wrapped intervals map to the open arcs containing S^1 . Thus $f^{-1}: \overline{\mathbb{R}} \rightarrow S^1$ is also an open map. Together with f being an open map, this proves that f is a homeomorphism. $\square$